MATH 3042- Assignment 4- Solutions

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Please write down the filled in R code you use to look up each of the probabilities.

Question 1: (4 marks)

A cloud storage system performs nightly backups of 80 databases. Each backup has a 97% probability of completing successfully without errors.

a) What is the probability that exactly 75 backups succeed?

b) What is the probability that at least 78 backups succeed?

c) Calculate the expected number of successful backups and the standard deviation.

d) Identify all the unusually low number of successful backups. Use the $Z < -2$ criteria

Let X be the number of successful backups. $X \sim Binom(80, 0.97)$

$a)\ P(x = 75) = C(80,75)(0.97)^{75}(0.03)^{5} = dbinom(75, 80, 0.97) = 0.0595$

$b)\ P(x \geq 78) = 1 - P(x \leq 77) = 1 - pbinom(77, 80, 0.97) = 0.56812$

$c) E[X] = \mu = np = 80 \times 0.97 = 77.6$

$\sigma = \sqrt{npq} = \sqrt{80 \times 0.97 \times 0.03} = 1.525778$

$d) -2 = \frac{x - 77.6}{1.525778}$ solve for $x = 74.5$

Less than or equal to 74 $(x = 0, 1, 2, \ldots, 74)$ successful backups is unusually low.

Question 2: (4 marks)

A REST API has 40 endpoints, of which 12 contain security vulnerabilities. A penetration tester randomly selects 8 endpoints to examine.

- a) What is the probability that exactly 3 vulnerable endpoints are selected?
- b) What is the probability that at least 2 vulnerable endpoints are selected?
- c) Calculate the expected number of vulnerable endpoints in the sample.

Hypergeometric Distribution

a) $P(x = 3) = \frac{C(12,3) \times C(28,5)}{C(40,8)}$ = dhyper(3,12,28,8)= 0.281148

b) $P(x \geq 2) = 1 - phyper(1,12,28,8) = 0.7748$

c) $E[X] = 8 \times \left(\frac{12}{40}\right) = 2.4$, $\sigma^2 = 2.4 \times \left(1 - \frac{12}{40}\right) \times \left(\frac{40-8}{39}\right) = 1.38, \qquad \sigma = 1.17$

Question 3: (4 marks)

A genetic algorithm has a 15% chance of finding an optimal solution on each iteration.

- a) What is the probability that the first optimal solution is found on the 8th iteration?
- b) What is the expected number of iterations needed to find the first optimal solution?
- c) Calculate the standard deviation of the number of iterations.

Let X be the number of trials till the first success: $x = 1,2,3, \ldots$

$X \sim Geom(p = 15\%)$

a) $P(x = 8) = (0.85)^7(0.15) = dgeom(7, 0.15) = 0.048$

b) $E[X] = \frac{1}{0.15} = 6.7$ we expect 7 iterations till first success

c) $\sigma^2 = \frac{0.85}{0.15^2} = 37.8$, $\sigma = 6.1$

Question 4: (4 marks) A monitoring system detects an average of 5 errors per hour in a distributed application.

- a) What is the probability of detecting exactly 7 errors in a given hour?
- b) What is the probability of detecting at most 3 errors in a given hour?
- c) What is the probability of detecting more than 8 errors in a given 3-hour time interval?

Let X be the number of errors $X \sim Poisson\ (\mu = 5\ per\ hour)$

a) $P(x = 7) = dpois(7, 5) = 0.104$
b) $P(x \leq 3) = ppois(3, 5) = 0.265$
c) $P(x > 8) = 1 - ppois(8, 3 \times 5) = 0.963$

Q5-
A: Poisson. Average number of events per unit of time is given and the events are independent
B: Geometric, independent trials, fixed probability of success. Random variable: number of trials till success
C: Hypergeometric: sampling without replacement from a finite population
D: Binomial. Independent trials, fixed probability of success, known number of trials. Random variable: number of success.

Question 5: (4 marks)

For each scenario, identify which probability distribution is most appropriate (Binomial, Hypergeometric, Geometric, or Poisson) and explain why. No need to solve the problem, just identify the appropriate distribution.

- a) A server crashes randomly with an average of 2 crashes per month. You want to find the probability of exactly 3 crashes next month.
- b) You flip a biased coin (70% heads) repeatedly until you get the first tails. You want to know the probability it takes exactly 5 flips.
- c) A dataset contains 200 records, 45 of which have missing values. You randomly sample 20 records and want to know the probability that exactly 5 have missing values.
- d) You run 100 unit tests, each with a 95% pass rate. You want the probability that exactly 97 tests pass.